

E-commerce Sales SQL Project - Subquery Questions

This section of the project demonstrates the use of SQL subqueries. Subqueries allow us to nest queries inside other queries for more complex analysis.

Find customers who placed at least one order in North region

Query:

```
SELECT DISTINCT customer_id FROM orders WHERE customer_id IN ( SELECT  
customer_id FROM orders WHERE region = 'North' );
```

Find products that have a price higher than the average price

Query:

```
SELECT product_id, price FROM orders WHERE price > ( SELECT AVG(price)  
FROM orders );
```

Find orders that have revenue greater than the smallest order revenue

Query:

```
SELECT order_id, revenue FROM orders WHERE revenue > ( SELECT  
MIN(revenue) FROM orders );
```

Find categories that were sold in more than one region

Query:

```
SELECT DISTINCT category FROM orders o WHERE ( SELECT COUNT(DISTINCT  
region) FROM orders WHERE category = o.category ) > 1;
```

Find customers who spent the maximum total revenue

Query:

```
SELECT customer_id, SUM(revenue) AS total_spent FROM orders GROUP BY  
customer_id HAVING SUM(revenue) = ( SELECT MAX(total_spent) FROM ( SELECT  
customer_id, SUM(revenue) AS total_spent FROM orders GROUP BY customer_id  
) AS sub );
```